Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	UMvengers
Project / Solution Name	UMvengers-Orbyte
Name (Leader)	Wong Hanz
Email & Phone (Leader)	wonghanz2007@gmail.com 01159470337
University Affiliation(s)	University Malaya
Live Prototype URL	https://u-mvengers-orbyte-runner.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/AlyssaWoe/UMvengers-Orbyte.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

UMvengers-Orbyte connects a web-based dashboard, backend workflows, AI services, and a database to automate SME payment follow-ups. Users submit invoice information through the interface, where Docling supports document extraction and Pydantic validates the data before it is stored in PostgreSQL.

The Payment Surveillance Agent monitors due dates and overdue invoices. Laya supports payment-situation classification, while the DeepSeek API helps interpret customer replies and draft suitable follow-up messages. LangGraph coordinates the workflow, including business-rule checks and human approval for uncertain, disputed, or high-value cases. Payment outcomes can then be recorded for reconciliation and audit tracking.

The web application is deployed on Render, providing access to the prototype. Together, these components form a connected workflow from invoice intake to payment monitoring, AI-assisted follow-up, approval, and record updates. The architecture is modular, allowing individual components to be tested and improved independently. Any integrations that are simulated or not yet implemented will be clearly identified during the Phase 2 demonstration.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python	Backend logic and workflow integration	PSF License
JavaScript	Frontend interactions	ECMAScript standard
Docling	Invoice and document extraction	MIT
Pydantic	Invoice data validation	MIT
LangGraph	AI-agent workflow orchestration	MIT

Name & Version	Purpose in Project	License
PostgreSQL	Invoice and payment data storage	PostgreSQL License
DeepSeek API	Customer reply understanding and message drafting	Commercial API; governed by provider terms
Laya	Payment classification and decision support	Verify the specific implementation’s license or service terms
Render	Cloud hosting and deployment	Proprietary hosting service; provider terms

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek API	Understand customer replies and draft payment reminders	Free
Laya	Classify payment situations and support decision-making	Free
Docling	Extract invoice information from uploaded documents	Free
PostgreSQL	Store invoice, payment and follow-up records	Free
Render	Host and deploy the Orbyte web application	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek-V4.1-Flash *(or the exact model name you used)*	DeepSeek	Pre-trained	DeepSeek API Terms of Service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ ***Yes*** ✅ ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit	Web hosting	Major (as requested)
ChatGPT	Construct the documentation	Minor
Gemini Nano Banana	Generating and refining presentation visuals and slide assets	Moderate
Antigravity	Designing presentation slides and visual layouts	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used generative AI to improve productivity in documentation, presentation design, and visual content creation. All AI-generated text, images, and slide designs were reviewed, edited, and verified by team members to ensure technical accuracy, consistency, and alignment with project objectives. Visual assets were checked for relevance and originality, while documentation was validated against the implemented system. The team remained fully responsible for all final deliverables, design decisions, and technical implementation.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ ***Yes*** ✅ ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Web Platform	Hosting purpose.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Yes, as it might be replaced with our own web domain for cloud hosting.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application processes business data required for payment follow-up, including customer names, invoice details, payment status, communication history, and company knowledge documents uploaded to the LLM Wiki and Unified Data Source. AI requests are sent to the DeepSeek API solely to generate responses or recommendations. Personally identifiable information is minimised where possible, and only the necessary context is retrieved through the RAG pipeline. Data is protected using authenticated access controls, HTTPS encryption during transmission, and role-based permissions. No payment credentials or financial account information are collected. Project data is retained only for operational purposes and may be deleted upon request or according to the organisation's data retention policy.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The solution inherits limitations common to large language models, including hallucinations, incomplete reasoning, and dependence on the quality of retrieved knowledge. AI-generated recommendations may be inaccurate if company documents are outdated or incomplete. The system also depends on the availability and rate limits of the DeepSeek API. To mitigate these risks, the platform uses Retrieval-Augmented Generation (RAG) to ground responses in trusted company knowledge, while a compliance skill enforces predefined business and government guidelines before automated actions are taken. Human review is maintained for critical decisions and customer communications. Laya coordinates specialised AI agents to minimise redundant model calls, improving efficiency and reducing operational costs while providing a fallback path if a subagent fails.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document. ✅
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required. ✅
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document. ✅
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team. ✅
- ☐ Any use of official SDC sponsor resources has been declared in Section 5. ✅
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.✅
- ☐ All fields in this template have been completed within the word limits stated.✅

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

WONG HANZ

24/10/2026